

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation

Anonymous WACV Datasets Track submission

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Abstract

The ABSTRACT is to be in fully justified italicized text, at the top of the left-hand column, below the author and affiliation information. Use the word “Abstract” as the title, in 12-point Times, boldface type, centered relative to the column, initially capitalized. The abstract is to be in 10-point, single-spaced type. Leave two blank lines after the Abstract, then begin the main text. Look at previous WACV abstracts to get a feel for style and length.

Table 1. L-shaped protocol. The 3D UX-Net row tests cross-dataset consistency; the BTCV column tests cross-architecture consistency; they meet at the UX-Net/BTCV anchor (*, completed). Off-axis cells (–) are not run.

Backbone	BTCV (CT)	FeTA (MRI)	Hepatic Vessel (CT)
3D UX-Net	✓*	✓	✓
Swin UNETR	✓	–	–
MedNeXt-M-K3	✓	–	–

1. Experimental Setup

Experiment plan. Our evaluation study reproduces EfiDec3D [11] and evaluates it with an *L-shaped* protocol (Table 1) that separately tests cross-dataset and cross-architecture consistency without an expensive full factorial. A **dataset axis** runs the matched 3D UX-Net [8] pair on three diverse tasks from the EfiDec3D evaluation—BTCV abdominal CT, FeTA [10] fetal-brain MRI, and MSD Task08 Hepatic Vessel [1] (CT, thin vessels/small lesions)—and an **architecture axis** runs matched pairs on BTCV for three backbone families: 3D UX-Net (large-kernel CNN), Swin UNETR [13] (hierarchical Transformer), and MedNeXt-M-K3 [12] (ConvNeXt-style CNN). The two axes share the 3D UX-Net/BTCV anchor, which also carries the mechanism analysis below.

Dataset and split. The primary task is the 13-organ BTCV/Synapse abdominal CT benchmark with 18 training and 12 validation volumes: spleen, right and left kidney, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, portal and splenic veins, pancreas, and right and left adrenal glands. Images are intensity-windowed and resampled following the public EfiDec3D pipeline; the same validation subjects are paired across all model comparisons on a given dataset.

Two comparison regimes. Within each cell the full and EfiDec3D models share the same encoder *architecture* and

differ in the decoder (channels reduced to the minimum encoder width, additive skip aggregation, resolution factor two). We analyze them under two regimes. (i) **End-to-end (ecological)**. Both models are trained independently, so they share the encoder architecture but not its weights; their flips reflect the deployed full-vs-efficient difference but conflate decoder capacity with encoder-weight and optimization differences. (ii) **Frozen shared encoder (causal)**. We take one trained encoder, freeze it, and train the decoder variants on *identical* features, so prediction differences are attributable to the decoder alone. Both regimes hold the encoder architecture fixed, so the study characterizes the decoder and draws no conclusions about the encoder itself.

Frozen-encoder factor decomposition. EfiDec3D removes decoder computation in two ways at once—narrower channels and a dropped high-resolution stage. To attribute effects, we decompose them with a 2×2 factorial on the frozen encoder, all within the EfiDec3D decoder family so only the two knobs vary: full (highest-resolution stage, wide channels), channel-only (wide $\rightarrow$ 48 channels), resolution-only (drop the high-resolution stage), and the combined EfiDec3D decoder. Pairwise flips (Full $\rightarrow$ channel-only, Full $\rightarrow$ resolution-only, Full $\rightarrow$ combined) isolate which factor changes which regions. The frozen encoder is optimized for the full decoder, so “efficient still matches on frozen features” is a *conservative* redundancy result.

062 Null-pair seed control. Because the net flip is tiny, we
063 bound seed noise with a same-architecture null pair: two
064 Full models trained with different (predeclared) seeds. The
065 claim is that Full-vs-EffiDec3D net flips and their concen-
066 tration exceed this Full-vs-Full null.

067 Implementation. We reproduce the EfficDec3D training
068 protocol on a single NVIDIA RTX 5090 (32 GB, Black-
069 well) under PyTorch 2.6 / CUDA 12.8 with MONAI, us-
070 ing BF16 mixed precision (no loss scaling). Each model
071 is trained for 45,000 optimizer steps (validation every 250)
072 with AdamW at learning rate 10^{-3} , a Dice-cross-entropy
073 loss, four 96^3 random-crop samples per volume, and full
074 in-memory caching. Sliding-window inference uses a 96^3
075 window, batch size four, overlap 0.7, and constant blending.
076 We evaluate one full run and two independently initialized
077 EfficDec3D runs per cell to expose seed sensitivity. Seg-
078 mentation quality is reported with Dice and 95th-percentile
079 Hausdorff distance (HD95), and efficiency with parameters,
080 multiply-accumulate operations (MACs), latency, and peak
081 memory.

082 Core quantities. Let $\hat{y}_E(v)$, $\hat{y}_F(v)$, and $y(v)$ be the ef-
083 ficient prediction, full prediction, and ground-truth label at
084 voxel v . We measure where the full decoder changes predic-
085 tions with *prediction flips*, the standard tool for comparing
086 two classifiers on paired data [2, 14]:

$$\mathbf{087} \quad P(v) = \mathbb{I}[\hat{y}_E(v) \neq y(v) \wedge \hat{y}_F(v) = y(v)], \quad (1)$$

$$\mathbf{088} \quad N(v) = \mathbb{I}[\hat{y}_E(v) = y(v) \wedge \hat{y}_F(v) \neq y(v)], \quad (2)$$

$$\mathbf{089} \quad U(v) = P(v) - N(v), \quad (3)$$

090 a *positive* flip (full corrects efficient), a *negative* flip (full
091 degrades efficient), and the *net* flip. P and N are ex-
092 actly the discordant cells of the paired-agreement table be-
093 hind McNemar’s classifier-comparison test [3, 9]; volume
094 rates are $R_{\text{pos}} = \text{mean}_v P(v)$, $R_{\text{neg}} = \text{mean}_v N(v)$, and
095 $R_{\text{net}} = R_{\text{pos}} - R_{\text{neg}}$. As test-time signals computable
096 from the efficient model alone we use predictive entropy
097 $H(v) = -\sum_c p_c(v) \log p_c(v)$, confidence $\max_c p_c(v)$, and
098 Monte-Carlo dropout variance [4].

099 Characterization metrics. We group the analysis into
100 three questions; each metric is computed per grid cell and
101 aggregated over subjects with a bootstrap, since the twelve
102 validation volumes—not the correlated voxels—are the in-
103 dependent units.

104 (1) Heterogeneity: is the benefit evenly distributed?

- 105 • O1, boundary error ratio.** Mean voxel error on the
106 eroded boundary band divided by the eroded-interior er-
107 ror; >1 indicates boundary-concentrated error.
- 108 • O4, per-organ difficulty.** Per-class Dice and mean en-
109 tropy H , ranking the structures the decoder struggles on.

- O10, size-difficulty relation.** Spearman ρ between organ
 volume and difficulty, and the partial correlation of H and
 error given log-volume.

- Boundary-resolved flips.** Positive/negative/net flip
 rate binned by Euclidean distance to the nearest GT
 boundary—testing whether the decoder *benefit* (not just
 error) concentrates near boundaries. Complemented by
 Normalized Surface Dice (NSD) at 1–2 voxel tolerance
 for a surface-level view.

**(2) Concentration: does a small subset carry most net ben-
 efit?**

- O2, entropy sparsity.** $\Pr[H(v) > \tau]$, the fraction of
 high-entropy voxels.

- O5, uncertainty-benefit correlation.** Subject-level
 Pearson $r(H, U)$ between binned entropy and net flips.

- O9^{orc}, voxel recovery.** $\sum_{\text{top-}k\%} P(v) / \sum P(v)$, the
 share of positive flips in the highest-entropy $k\%$ of
 voxels—a non-deployable upper bound.

- O6, training persistence.** Mean entropy across check-
 points, testing whether concentration is a training tran-
 sient.

(3) Predictability: can a test-time signal find the regions?

- O3, flip discrimination and calibration.** AUROC and
 AUPRC of entropy predicting flips [7], and expected cal-
 ibration error (ECE) [6].

- O9, opportunity curve.** A risk-coverage curve [5]
 over contiguous 16^3 blocks (plus a 4-voxel halo) ranked
 by entropy, reporting net-flip utility recovered versus a
 matched-random selector.

- O11, routing-signal comparison.** Subject-level corre-
 lation of entropy, confidence, and MC-dropout with net
 flips.

Cross-dataset (O7) and cross-backbone (O8) consistency
 re-evaluate these metrics across the rows and columns of
 Table 1.

[TODO: Completed results below are the 3D UX-
 Net/BTCV grid cell. Remaining cells (other backbones
 and datasets) are in progress; cross-backbone (columns) and
 cross-dataset (rows) consistency are reported once enough
 cells are filled.]

2. Results

2.1. Controlled Reimplementation

We reproduce the matched Full-UX/Effi-UX pair and report
 its fidelity per dataset (Table 2). On BTCV our Effic-UX
 reaches 0.773 mean Dice over two seeds, 1.7–2.2 points
 below the published 0.793, while Full-UX is within 0.6
 points of its published value; the residual gap reflects our re-
 constructed identity-affine data rather than the paper’s pro-
 cessed set, and a FP32-versus-BF16 check ruled out a nu-
 merical cause. Efficiency reproduces the published reduc-

Table 2. Reproduction fidelity of the matched 3D UX-Net pair, per dataset. We report mean Dice of our runs against the published EffiDec3D value; Δ is our Effi-UX minus the published Effi-UX. Data were reconstructed with identity affine metadata, so this is a controlled reimplement, not an exact reproduction.

Dataset	Full-UX	Effi-UX	pub. Effi-UX
BTCV (CT, 13-organ)	.792	.773	.793
FeTA (MRI)	—	—	—
HepaticVessel (CT)	—	—	—

[TODO: Fill FeTA/HepaticVessel rows as the dataset-axis cells complete.]

Table 3. Prediction-flip rates across efficient-decoder seeds (fractions of evaluated voxels). Positive/negative flips are the discordant cells of the paired comparison [9, 14].

Metric	seed 0	seed
Positive flip rate R_{pos}	.00339	.0029
Negative flip rate R_{neg}	.00209	.0022
Net flip rate R_{net}	.00130	.0006
Net flip rate, subject 95% CI	[TODO: re-run]	[TODO: re-run]

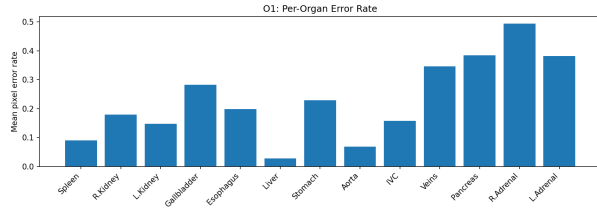


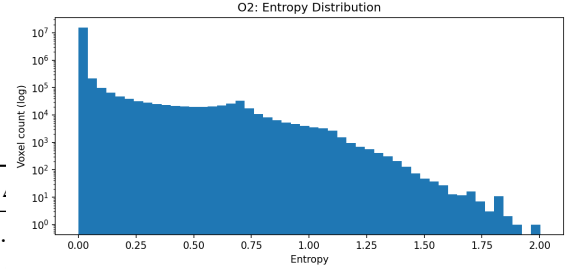
Figure 1. Per-organ error rate for Effi-UX (seed 1). Difficulty is strongly uneven across anatomy.

tions: Effi-UX uses $14.1 \times$ fewer MACs, $17.9 \times$ fewer parameters, $\sim 5 \times$ lower latency, and $\sim 7.7 \times$ lower peak memory. Treating these matched models as fixed, the rest of the paper characterizes *where* the full decoder’s extra computation changes predictions—the question EffiDec3D’s aggregate evaluation leaves open.

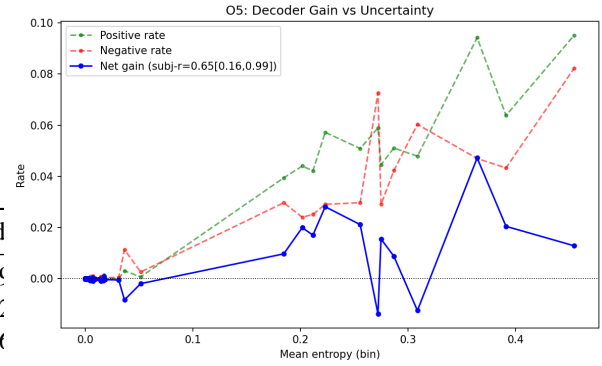
2.2. Heterogeneity Metrics

The full decoder’s benefit is small and *bidirectional*: Full-UX corrects 0.290% of Effi-UX’s voxels (R_{pos}) while degrading 0.225% (R_{neg}) for seed 1, and 0.339%/0.209% for seed 0, so the net flip rate R_{net} is only 0.065–0.130% of voxels (Table 3). Reporting positive flips alone would overstate the decoder’s benefit roughly twofold.

This small net benefit is far from uniformly distributed. Error concentrates at structure boundaries: mean boundary error is 0.1892 versus 0.0481 in the eroded interior, a $3.93 \times$ ratio [TODO: add subject 95% CI from re-run]. It is also



(a) Voxel entropy (log-count).



(b) Net flips vs. entropy.

Figure 2. Concentration of difficulty and net flips (seed 1). Uncertainty is spatially sparse (a), and net flips rise with it (b).

organ-dependent (Fig. 1), with the adrenal glands, pancreas, and veins hardest, and it scales with structure size: the organ-size/difficulty Spearman correlation is $\rho = -0.54$, and entropy still explains organ difficulty after controlling for log volume (partial $r = 0.81$). Decoder benefit is therefore heterogeneous across region, organ, and scale. Crucially, resolving flips by distance to the GT boundary shows the *net flip* rate itself—not just error—rises toward the boundary [TODO: per-bin net flips + NSD from re-run], and this effect exceeds a same-architecture null pair (Full seed 0 vs. seed 1) [TODO: null-pair floor], ruling out seed noise.

2.3. Concentration Metrics

A small, high-uncertainty subset carries most of the net benefit. Predictive entropy is spatially sparse: the median voxel entropy is near zero and only 1.18% of voxels exceed 0.5 (Fig. 2a). Net flips concentrate in exactly this tail—the subject-level Pearson correlation between binned entropy and net flips is $r = 0.646$ (95% CI [0.158, 0.994]) for seed 1 and $r = 0.665$ ([0.171, 0.996]) for seed 0, both bounded above zero (Fig. 2b). A voxel-level oracle makes the concentration explicit: ranking foreground voxels by entropy, the top 20% contain 86.4% of all positive flips in both seeds. This concentration is not a transient of early training: mean entropy falls from 0.0279 at 5k iterations to 0.0104 at 45k,

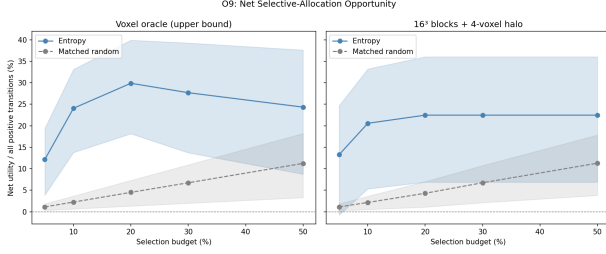


Figure 3. Region-level opportunity (seed 1). Contiguous 16^3 blocks ranked by entropy recover far more *net* flip utility than a matched-random selector at every budget $\geq 10\%$; the left panel is a non-deployable voxel oracle.

yet the sparse difficult residual persists.

2.4. Predictability Metrics

Concentration is only actionable if the beneficial regions can be found without the ground truth. We assess this as an error-discrimination problem: predictive entropy should separate flip voxels from the rest [7]. Per-subject AUROC and AUPRC of entropy predicting error, together with confidence calibration (ECE) [6], are reported with a subject bootstrap [TODO: fill AUROC/AUPRC/ECE + CI from re-run; the pooled-bin entropy-error correlation $r = 0.97$ is retained only as a descriptive diagnostic].

We then convert this into a deployable, selective-prediction analysis [5]. Non-overlapping 16^3 blocks are ranked by mean entropy and expanded by a 4-voxel halo; we measure net flip utility recovered against a matched-random selector under a paired subject bootstrap (Fig. 3). At the predeclared 20% block budget—executing 29.1% of the foreground-union volume after halo expansion—entropy-ranked regions attain a net utility of 0.225 versus 0.043 for random, a roughly fivefold gain, with a paired 95% lower bound of $0.056 > 0$; the advantage holds at every budget $\geq 10\%$ and vanishes only at 5%. These regions capture all positive flips but also 77.5% of the negative ones, so the *net* gain, not positive recovery, is the honest headline.

Finally, we compare cheap routing signals against held-out net flips per subject. Entropy and confidence are statistically indistinguishable [TODO: fill subject-level correlations + paired CI from re-run; pooled values are 0.655 and 0.663], and Monte-Carlo dropout is invalid for this checkpoint because the architecture has no active dropout (its stochastic variance is effectively zero). We therefore retain confidence as a required baseline and do not claim entropy is uniquely optimal.

2.5. Mechanism: Which Factor Is Removable

The comparisons above use end-to-end models, which share the encoder architecture but not its weights. As a further, causally cleaner check, we *freeze* one trained encoder

and train the 2×2 decoder factorial on identical features, then compare Full→channel-only, Full→resolution-only, and Full→combined EffiDec3D [TODO: report per-factor flip rates and boundary profiles from the frozen-encoder run; hypothesis to test: dropping the high-resolution stage yields flips concentrated nearer the boundary, while channel reduction is spatially diffuse and smaller, which would explain why both are removable with little aggregate Dice change]. Because the frozen encoder was optimized for the full decoder, a near-match by the efficient decoder on those features would be a *conservative* statement of redundancy. We present this regime as validation of the decoder attribution rather than as the primary result.

References

- [1] Michela Antonelli, Annika Reinke, Spyridon Bakas, et al. The medical segmentation decathlon. *Nature Communications*, 13(1):4128, 2022. 1
- [2] Quentin Cormier, Mahdi Milani Fard, Kevin Canini, and Maya R. Gupta. Launch and iterate: Reducing prediction churn. In *Advances in Neural Information Processing Systems*, 2016. 2
- [3] Thomas G. Dietterich. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7):1895–1923, 1998. 2
- [4] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning*, pages 1050–1059, 2016. 2
- [5] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017. 2, 4
- [6] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, pages 1321–1330, 2017. 2, 4
- [7] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017. 2, 4
- [8] Ho Hin Lee, Shunxing Bao, Yuankai Huo, and Bennett A. Landman. 3d ux-net: A large kernel volumetric convnet modernizing hierarchical transformer for medical image segmentation. In *International Conference on Learning Representations*, 2023. 1
- [9] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. 2, 3
- [10] Kelly Payette, Priscille de Dumast, Hamza Kebiri, et al. An automatic multi-tissue human fetal brain segmentation benchmark using the fetal tissue annotation dataset. *Scientific Data*, 8(1):167, 2021. 1
- [11] Md Mostafijur Rahman and Radu Marculescu. Effid3d: An optimized decoder for high-performance and efficient 3d medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10435–10444, 2025. 1

- 295 [12] Saikat Roy, Gregor Koehler, Constantin Ulrich, Michael
296 Baumgartner, Jens Petersen, Fabian Isensee, Paul F. Jaeger,
297 and Klaus H. Maier-Hein. MedNeXt: Transformer-driven
298 scaling of ConvNets for medical image segmentation. In
299 *International Conference on Medical Image Computing and*
300 *Computer-Assisted Intervention*, pages 405–415, 2023. [1](#)
- 301 [13] Yucheng Tang, Dong Yang, Wenqi Li, Holger R. Roth,
302 Bennett Landman, Daguang Xu, Vishwesh Nath, and Ali
303 Hatamizadeh. Self-supervised pre-training of swin trans-
304 formers for 3d medical image analysis. In *Proceedings of*
305 *the IEEE/CVF Conference on Computer Vision and Pattern*
306 *Recognition*, pages 20730–20740, 2022. [1](#)
- 307 [14] Sijie Yan, Yuanjun Xiong, Kaustav Kundu, Shuo Yang, Siqu
308 Deng, Meng Wang, Wei Xia, and Stefano Soatto. Positive-
309 congruent training: Towards regression-free model updates.
310 In *Proceedings of the IEEE/CVF Conference on Computer*
311 *Vision and Pattern Recognition*, pages 14299–14308, 2021.
312 [2, 3](#)